

Electromagnetic solvers for medium-size objects

The Finite-Difference Time-Domain - FDTD

(Yee 1966)

Finite-Difference Time-Domain

- Scope

The FDTD technique offers many advantages as an electromagnetic tool for modeling, simulation and analysis. Its capabilities include:

- Broadband response centered about system resonances
- Handling arbitrary three-dimensional (3D) geometries including wires and thin metal sheets
- Interaction with scatterer(s) of any electrical conductivity (perfect or real conductors as well as lossless or real dielectric materials)
- Any medium inhomogeneity
- Any type of response, such as:
 - scattered field
 - map of currents, of power flux and absorption
 - field penetration depth
 - coupling between interior elements.

These capabilities hold for electromagnetic stimuli ranging from extremely low to extremely high frequencies. Systems responding to these stimuli are equally diverse. They can be small to large, inorganic as well as organic, in an exoatmospheric environment (plasma) to a subterranean one. Examples of types of systems that can be modeled by FDTD are:

- Shelters
- Aircrafts
- Humans
- Satellites
- Buried antennas

Time-Domain Maxwell Equations

$$\nabla \times \vec{E}(\vec{r}, t) = -\mu(\vec{r}) \frac{\partial}{\partial t} \vec{H}(\vec{r}, t) - \vec{M}(\vec{r}, t)$$

Faraday Law

$$\nabla \times \vec{H}(\vec{r}, t) = \varepsilon(\vec{r}) \frac{\partial}{\partial t} \vec{E}(\vec{r}, t) + \sigma(\vec{r}) \vec{E}(\vec{r}, t) + \vec{J}(\vec{r}, t)$$

Ampere Law

$$\nabla \cdot \vec{D}(\vec{r}, t) = \rho(\vec{r})$$

Gauss Law

$$\nabla \cdot \vec{B}(\vec{r}, t) = \rho_m(\vec{r})$$

Magnetic-flux conservation

$$\nabla \cdot \vec{J}(\vec{r}, t) = -\frac{\partial}{\partial t} \rho(\vec{r})$$

Charge conservation law

$$\vec{D}(\vec{r}, t) = \varepsilon(\vec{r}) \vec{E}(\vec{r}, t)$$

$$\vec{B}(\vec{r}, t) = \mu(\vec{r}) \vec{H}(\vec{r}, t)$$

$$\varepsilon(\hat{r}) \mu(\hat{r}) \sigma(\hat{r})$$

- Causal, time-invariant media
- Tensors for anisotropic media (plasma, ferrite)

Constitutive Equations

$$\vec{J}(\vec{r}, t)$$

Electric source (real or equivalent)

$$\vec{M}(\vec{r}, t)$$

Magnetic source (equivalent)

Time-Domain Maxwell Equations -contd

Curl-equations give a system of six (coupled) scalar differential equations of first order.

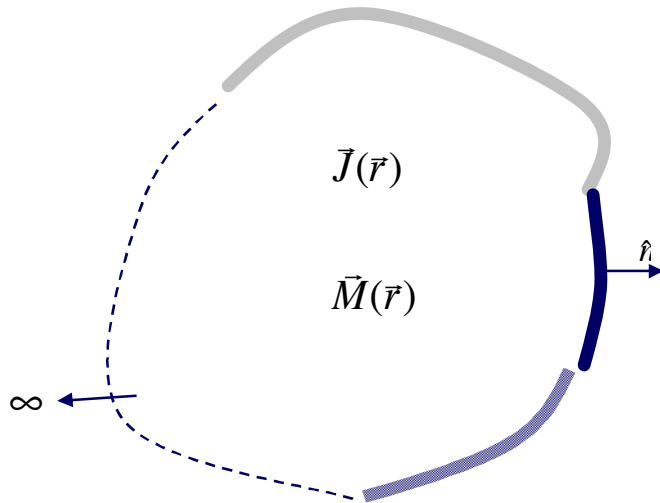
$$\frac{\partial E_x}{\partial t} = \frac{1}{\varepsilon} \cdot \left(\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} - \sigma \cdot E_x \right)$$

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \cdot \left(\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} \right)$$

- Space and time derivatives
- Boundary conditions

Time-Domain Maxwell Equations

-Boundary conditions



Radiation toward infinity

$$\lim_{r \rightarrow \infty} r [\vec{E}(\vec{r}) - \zeta \vec{H}(\vec{r}) \times \hat{r}] = 0 \quad \zeta = \sqrt{\mu / \epsilon}$$

 Electric boundary

$$\hat{n} \times \vec{E}(\hat{r}) = 0$$

 Magnetic Boundary

$$\hat{n} \times \vec{H}(\hat{r}) = 0$$

 Impedance boundary

$$\hat{n} \times \vec{E} \times \hat{n} = Z_s (\vec{H} \times \hat{n})$$

$$Z_s = \frac{1 + j}{\sigma \delta_s}$$

FDTD Approximation

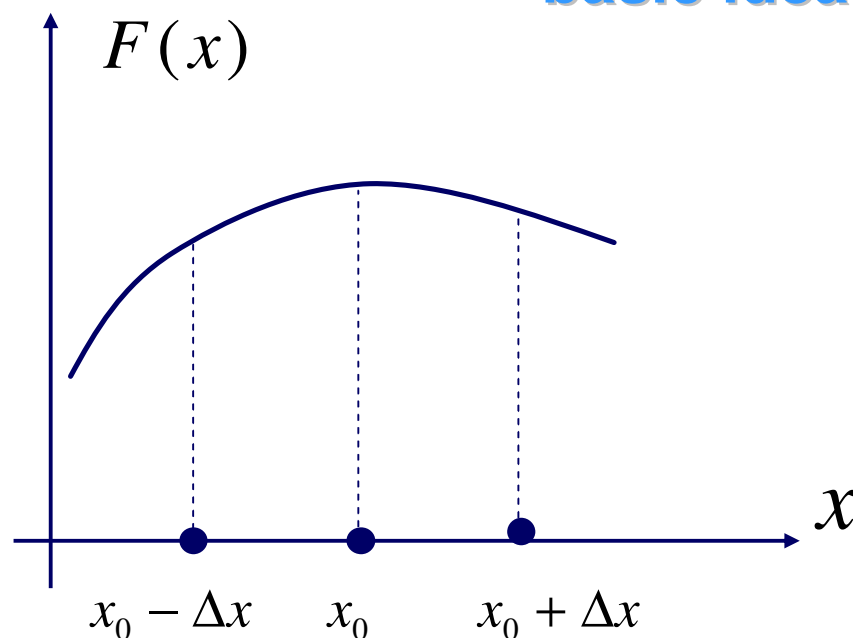
Partial differential equations will be approximated by replacing the time/space derivatives with finite differences

Continuous system → **Discrete system**

FDTD Approximation

- basic idea

The derivatives can be approximated by the incremental ratio



$$F'(x_0) = \begin{cases} \frac{F(x_0 + \Delta x) - F(x_0)}{\Delta x} & \text{Forward difference} \\ \frac{F(x_0) - F(x_0 - \Delta x)}{\Delta x} & \text{Backward difference} \\ \frac{F(x_0 + \Delta x) - F(x_0 - \Delta x)}{2\Delta x} & \text{Central difference} \end{cases}$$

Taylor expansion

$$F(x_0 + \Delta x) = F(x_0) + \Delta x F'(x_0) + \frac{\Delta x^2}{2} F''(x_0) + \frac{\Delta x^3}{6} F'''(\xi_1) \quad \xi_1 \in [x_0, x_0 + \Delta x]$$

$$F(x_0 - \Delta x) = F(x_0) - \Delta x F'(x_0) + \frac{\Delta x^2}{2} F''(x_0) - \frac{\Delta x^3}{6} F'''(\xi_2) \quad \xi_2 \in [x_0 - \Delta x, x_0]$$

Backward/Forward difference:

$$F'(x_0) = \frac{F(x_0 + \Delta x) - F(x_0)}{\Delta x} - \frac{\Delta x}{2} F''(x_0) - \frac{\Delta x^2}{6} F'''(\xi_1) \quad O(\Delta x)$$

Central difference:

$$F'(x_0) = \frac{F(x_0 + \Delta x) - F(x_0 - \Delta x)}{2\Delta x} - \frac{\Delta x^2}{6} F'''(\xi_3) \quad O(\Delta x^2)$$

$\xi_3 \in [x_0 - \Delta x, x_0 + \Delta x]$

Central-Difference approximation is second-order accurate

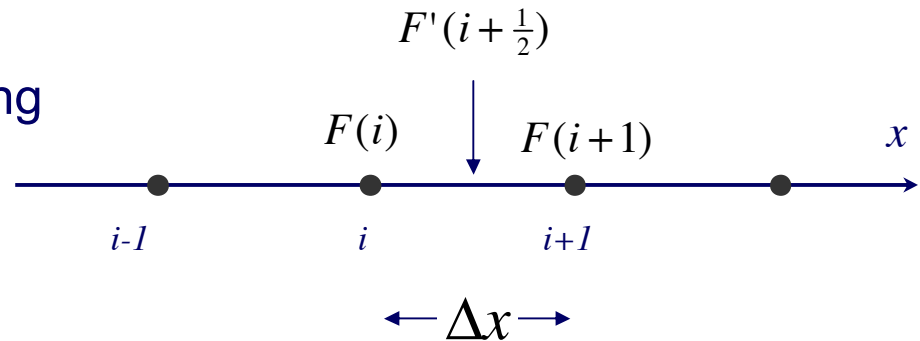
FDTD Approximation

- basic idea

Domain (space/time) sampling

$$x = i\Delta x$$

$$F(x) \equiv F(i)$$



A function and its first derivative are defined (sampled) at different points :

$$\frac{F(i+1) - F(i)}{\Delta x} \cong F'(i + \frac{1}{2})$$

In four dimensions:

$$F(x, y, z, t) = F(i\Delta x, j\Delta y, k\Delta z, n\Delta t) \equiv F^n(i, j, k)$$

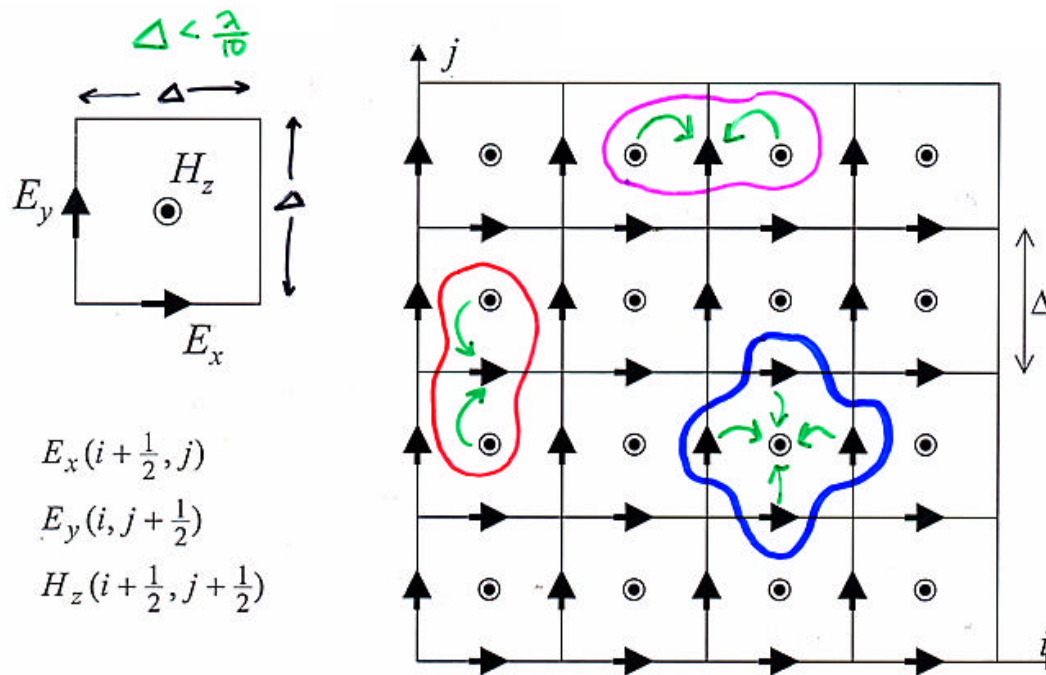
$$\left. \frac{\partial}{\partial x} F^n(i, j, k) \right|_{i+\frac{1}{2}} \neq \frac{F^n(i+1, j, k) - F^n(i, j, k)}{\Delta x}$$

$$\left. \frac{\partial}{\partial t} F^n(i, j, k) \right|_{n+\frac{1}{2}} \neq \frac{F^{n+1}(i, j, k) - F^n(i, j, k)}{\Delta t}$$

Space sampling grid

-two-dimensional case (E_x , E_y , H_z)

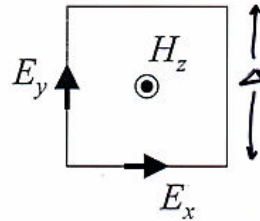
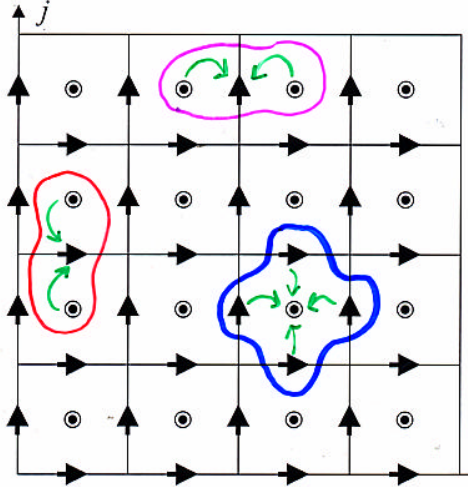
Yee cell



- Field samples are shifted in the elementary YEE cell
- **Cell-size:** $\Delta \leq \lambda/10$

Space sampling grid

-2D homogeneous case



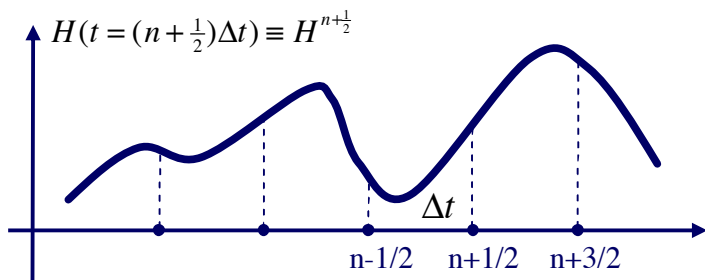
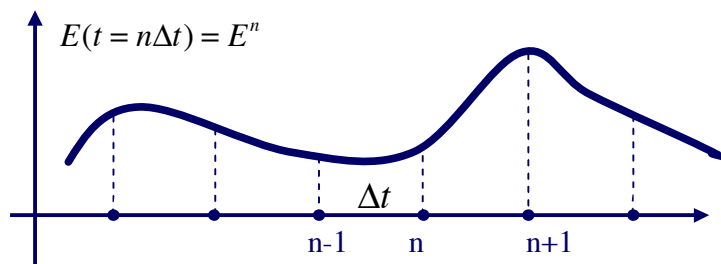
$$\epsilon \frac{\partial E_x}{\partial t} \left(i + \frac{1}{2}, j \right) = \frac{\partial H_z}{\partial y} \bigg|_{(i+\frac{1}{2}, j)} \cong \frac{H_z(i + \frac{1}{2}, j + \frac{1}{2}) - H_z(i + \frac{1}{2}, j - \frac{1}{2})}{\Delta}$$

$$\epsilon \frac{\partial E_y}{\partial t} \left(i, j + \frac{1}{2} \right) = - \frac{\partial H_z}{\partial x} \bigg|_{(i, j+\frac{1}{2})} \cong - \frac{H_z(i + \frac{1}{2}, j + \frac{1}{2}) - H_z(i - \frac{1}{2}, j + \frac{1}{2})}{\Delta}$$

$$\mu \frac{\partial H_z}{\partial t} \left(i + \frac{1}{2}, j + \frac{1}{2} \right) = \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} \bigg|_{(i+\frac{1}{2}, j+\frac{1}{2})}$$

$$\cong \frac{E_x(i + \frac{1}{2}, j + 1) - E_x(i + \frac{1}{2}, j)}{\Delta} - \frac{E_y(i + 1, j + \frac{1}{2}) - E_y(i, j + \frac{1}{2})}{\Delta}$$

Time sampling scheme



$$\Delta t \leq \frac{\Delta}{c\sqrt{\alpha}}$$

1D $\alpha=1$

2D $\alpha=2$

3D $\alpha=3$

Stability Condition
(Courant)

Electric and magnetic fields
are sampled at alternate times

$$\dot{E}_x \Rightarrow E_x^n(i + \frac{1}{2}, j) = E_x^{n-1}(i + \frac{1}{2}, j) + \frac{\Delta t}{\epsilon} \frac{H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}) - H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j - \frac{1}{2})}{\Delta y}$$

$$\dot{E}_y \Rightarrow E_y^n(i, j + \frac{1}{2}) = E_y^{n-1}(i, j + \frac{1}{2}) + \frac{\Delta t}{\epsilon} \frac{H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}) - H_z^{n-\frac{1}{2}}(i - \frac{1}{2}, j + \frac{1}{2})}{\Delta x}$$

$$\dot{H}_z \Rightarrow H_z^{n+\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}) = H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}) + \frac{\Delta t}{\mu} \frac{E_x^n(i + \frac{1}{2}, j + 1) - E_x^n(i + \frac{1}{2}, j)}{\Delta y} - \frac{\Delta t}{\mu} \frac{E_y^n(i + 1, j + \frac{1}{2}) - E_y^n(i, j + \frac{1}{2})}{\Delta x}$$

Complete 2D equations

Continuous case

$$\epsilon \left. \frac{\partial E_x}{\partial t} \right|_{t=(n-\frac{1}{2})\Delta t} = \epsilon \left. \frac{\partial H_z}{\partial y} \right|_{\substack{x=(i+\frac{1}{2})\Delta x \\ y=j\Delta y}} - \sigma E_x - J$$

FDTD approximation

$$E_x^n(i + \frac{1}{2}, j) = \left(\frac{1 - \frac{\sigma \Delta t}{2\epsilon}}{1 + \frac{\sigma \Delta t}{2\epsilon}} \right) E_x^{n-1}(i + \frac{1}{2}, j) +$$

$$\left(\frac{\Delta t / \epsilon}{1 + \frac{\sigma \Delta t}{2\epsilon}} \right) \frac{H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}) - H_z^{n-\frac{1}{2}}(i + \frac{1}{2}, j - \frac{1}{2})}{\Delta y} -$$

$$\left(\frac{\Delta t / \epsilon}{1 + \frac{\sigma \Delta t}{2\epsilon}} \right) J_x^n(i + \frac{1}{2}, j)$$

Flowchart

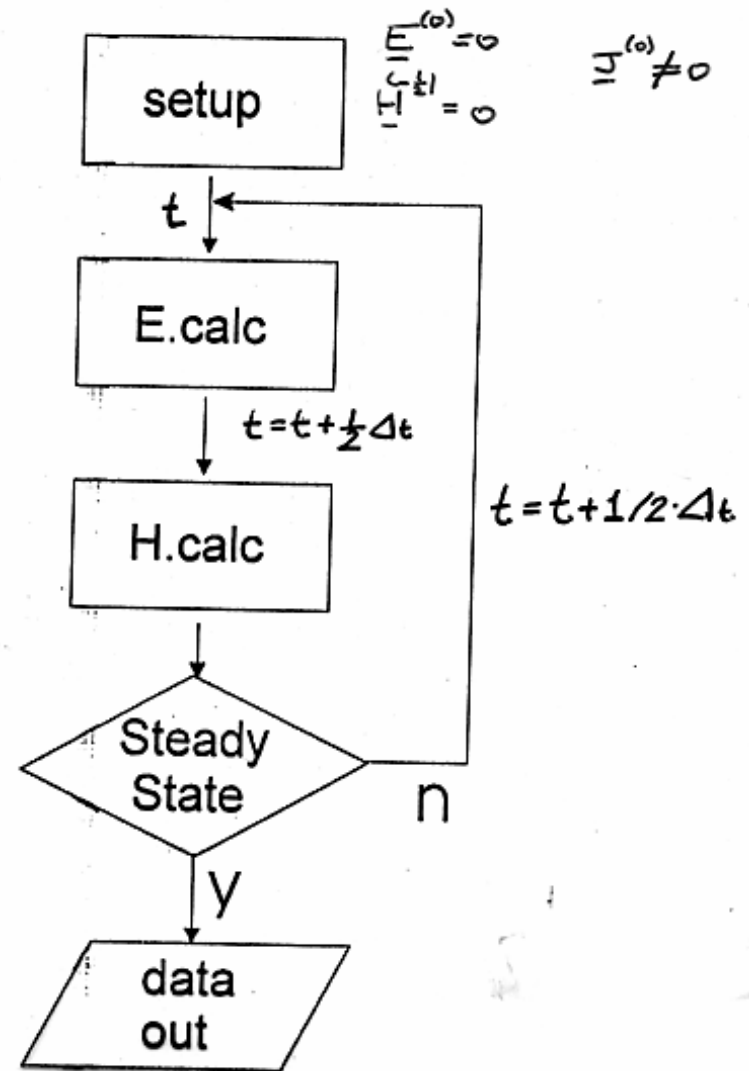
$$t = n\Delta t$$

$$E^{(n)} = f[E^{(n-1)}, H^{(n-0.5)}, J^{(n)}]$$

$$t = (n + 0.5)\Delta t$$

$$H^{(n+0.5)} = g[H^{(n-0.5)}, E^{(n)}]$$

- Explicit equations
- Iterative solution (updating)
- No matrix inversion
- Computational and memory complexity **proportional** to the number of cells (unknowns)



Three-dimensional scheme

Electric components are tangential to the cell edges

$$E_x^n(i + \frac{1}{2}, j, k)$$

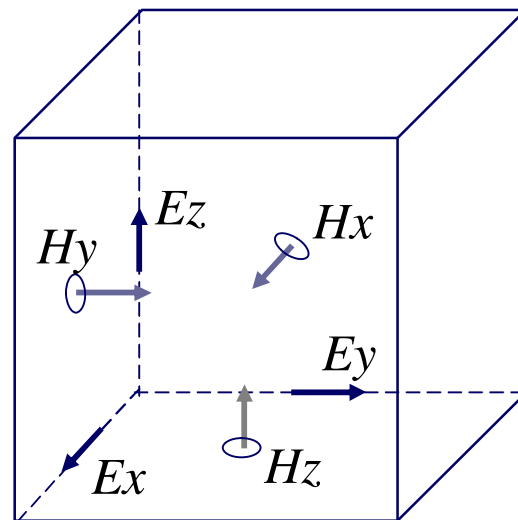
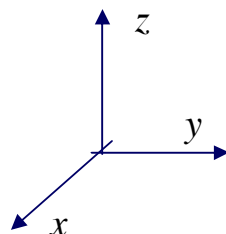
$$E_y^n(i, j + \frac{1}{2}, k)$$

$$E_z^n(i, j, k + \frac{1}{2})$$

$$H_x^{n+\frac{1}{2}}(i, j + \frac{1}{2}, k + \frac{1}{2})$$

$$H_y^{n+\frac{1}{2}}(i + \frac{1}{2}, j, k + \frac{1}{2})$$

$$H_z^{n+\frac{1}{2}}(i + \frac{1}{2}, j + \frac{1}{2}, k)$$



Magnetic components are orthogonal to the cell sides

Excitation waveforms

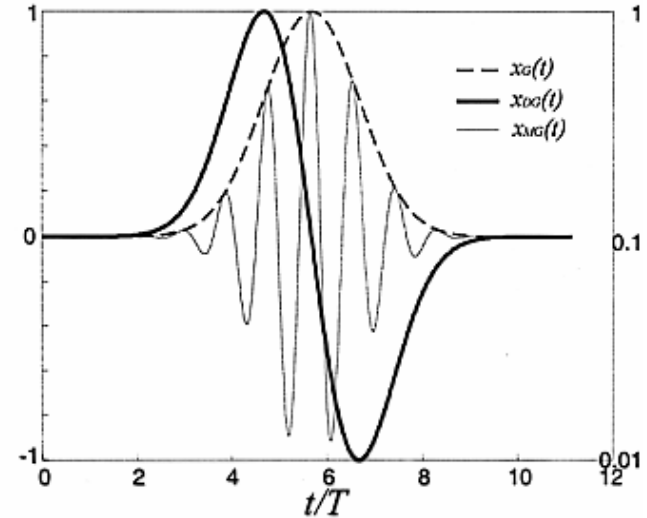
An e.m. source could have the following expression $J(\underline{r}, t) = L(\underline{r}) \cdot x(\underline{r})$

monochromatic

$$x_s(t) = \sin(2\pi f_0 t)$$

Sinusoid

$$f = f_0$$



wideband

$$x_G(t) = e^{-\frac{(t-\tau)^2}{2T^2}}$$

Gaussian pulse

$$0 < f \leq f_{\max}$$

$$x_{DG}(t) = -\frac{t-\tau}{T} e^{-\frac{(t-\tau)^2}{2T^2}}$$

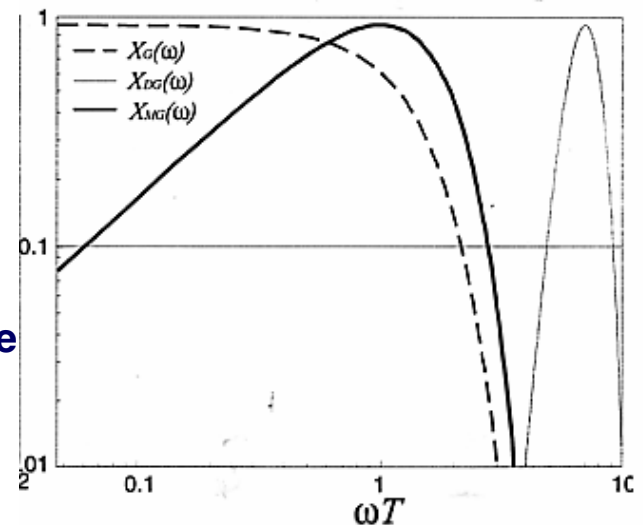
Derivated Gaussian pulse

$$f_{\min} < f \leq f_{\max}$$

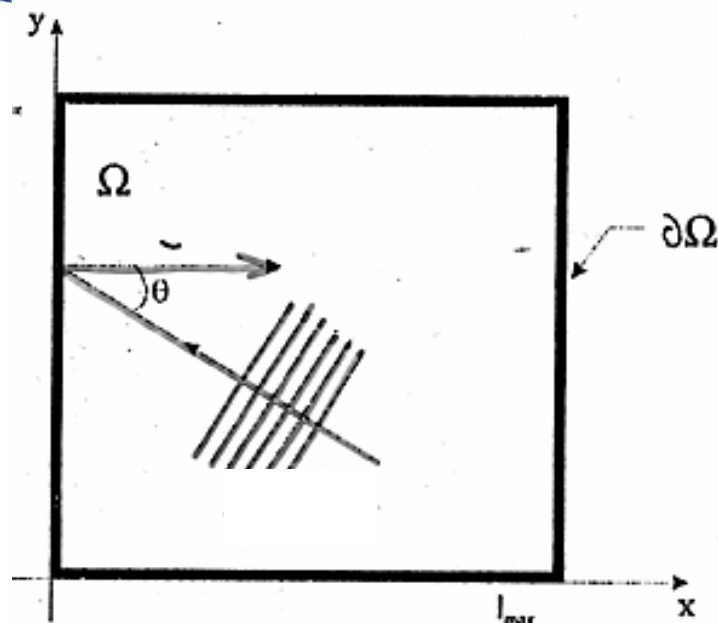
$$x_{MG}(t) = \sin(2\pi f_0 t) e^{-\frac{(t-\tau)^2}{2T^2}}$$

Modulated Gaussian pulse

$$|f - f_0| < \Delta f$$



Grid truncation



To be computational affordable, the simulation region should be truncated in the surrounding of the scatterers

Absorbing Boundary Conditions need to be enforced on the truncation sides to simulate the open space and avoid reflections

- One way equations (MUR)
- Perfectly matched layers (PML)

Absorbing Boundary Conditions

- One way equation

One Way Equation (Engquist & Majda, 1970)

A partial differential equation that permits wave propagation only in some directions. When applied to the **outer boundary** of an FDTD grid, a one-wave equation numerically **absorbs** impinging scattered waves.

Hypothesis: the field impinging the boundary is supposed to be as a plane wave

$$\frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2} - \frac{1}{c^2} \frac{\partial^2 U}{\partial t^2} = 0$$

2D wave equation

U: field component
parallel to the
boundary

$$L \equiv \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} - \frac{1}{c^2} \frac{\partial^2}{\partial t^2} \equiv D_x^2 + D_y^2 - \frac{1}{c^2} D_t^2$$

Partial
differential
operator

$$L = L^+ L^-, \quad L^\pm = D_x \pm \frac{D_t}{c} \sqrt{1 - S^2}$$

Factorization
for $x=0$ boundary

$$S = \frac{D_y}{D_t / c}$$

$$L^- U = 0$$

Applied to $x=0$ works as an exact analytical ABC for wave motion originating from the interior of spatial domain at any angle α

Absorbing Boundary Conditions - Perfectly Matched Layers

- anisotropic medium
- perfectly matched to the free-space
- increasing conductivity toward termination
- terminated on a PEC box

$$\underline{\underline{\epsilon}} = \epsilon_0 \underline{\underline{\epsilon_r}} \underline{\underline{s}}$$

$$\underline{\underline{\mu}} = \mu_0 \underline{\underline{\mu_r}} \underline{\underline{s}}$$

- PML parameters,

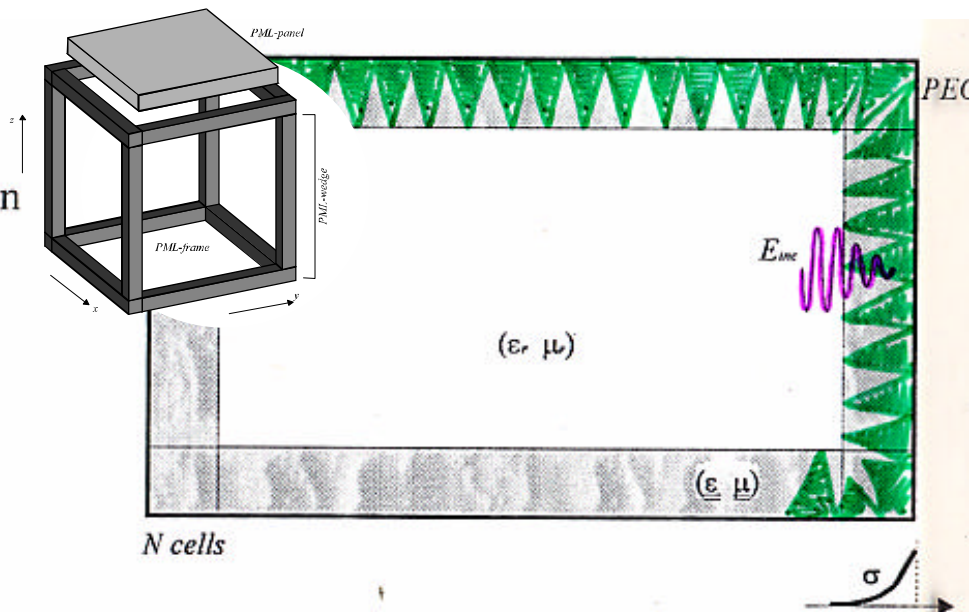
where

$$\underline{\underline{s}} = \begin{bmatrix} \frac{s_y s_z}{s_x} & 0 & 0 \\ 0 & \frac{s_x s_z}{s_y} & 0 \\ 0 & 0 & \frac{s_x s_y}{s_z} \end{bmatrix}$$

$$s_x = 1 + \frac{\sigma_x}{j\omega\epsilon_0}$$

$$s_y = 1 + \frac{\sigma_y}{j\omega\epsilon_0}$$

$$s_z = 1 + \frac{\sigma_z}{j\omega\epsilon_0}$$

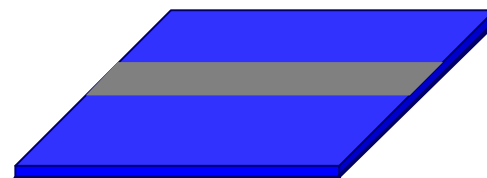
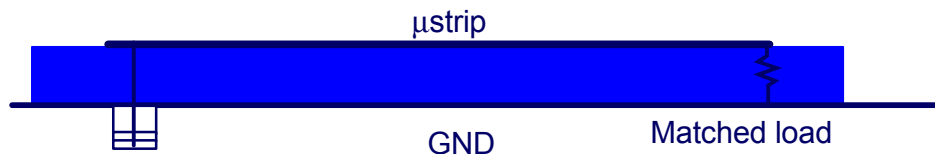


within PML, only field component perpendicular to layer sides must be updated according to special equations.

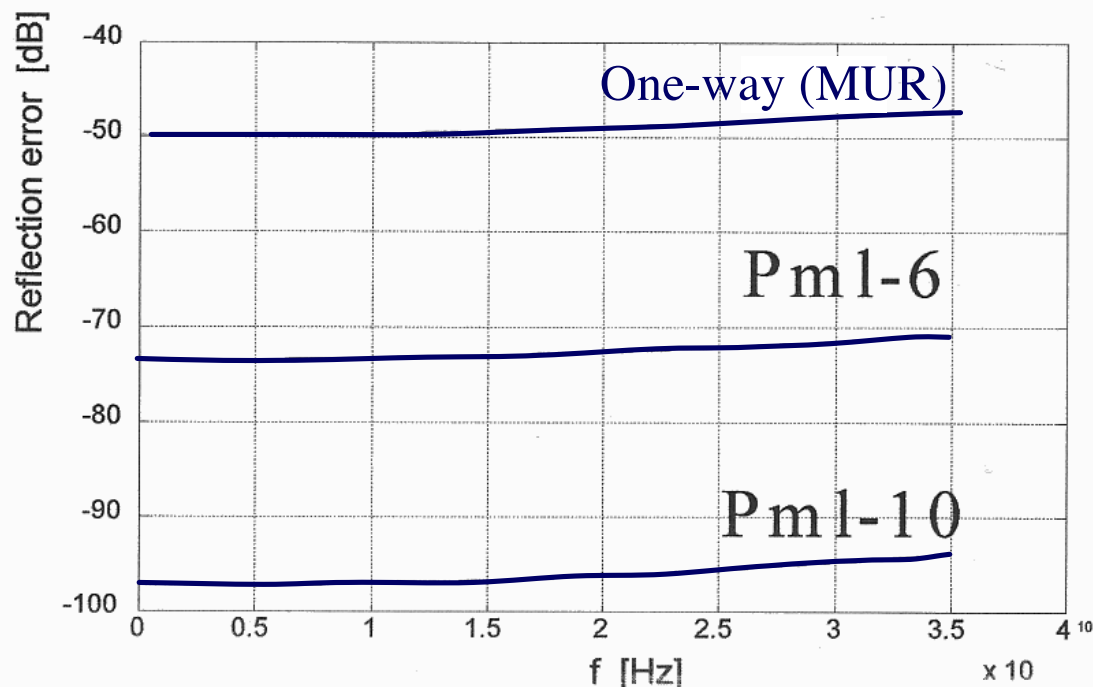
Reflection error < -60dB

Absorbing Boundary Conditions

- one-way eq. vs. PML



Reflection coefficient at the source port



Absorbing Boundary Conditions

- one-way eq. (Mur) vs. PML

One way (MUR 1st order)

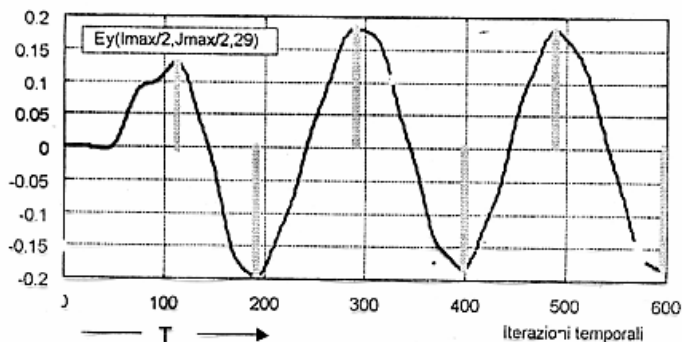
- enforced only on the boundary tangential field components (last cell)
- computationally light
- lower accuracy)
- useful for a (first time) fast analysis

PML

- enforced on a thick boundary shell (6-12 cells)
- computational hard (20% runtime overhead)
- high accuracy ($\Gamma < -70\text{dB}$)
- useful for refinement analysis

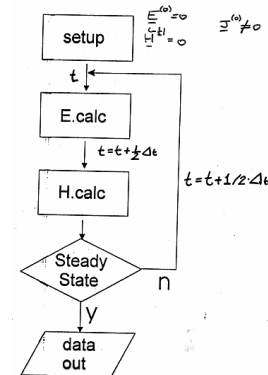
Time iterations progresses until the electromagnetic system exhibits the steady-state.

Monochromatic regime

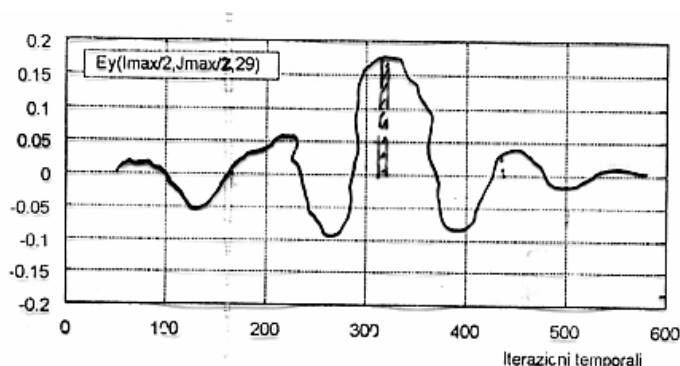


- After a transient each field sample oscillates at a same frequency
- Test over the maximum/minimum within successive periods

$$|E(i, j, k)_{\max} + E(i, j, k)_{\min}| < \delta$$

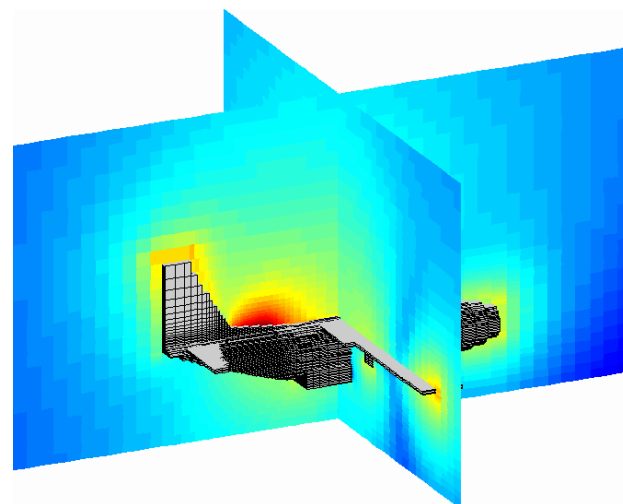
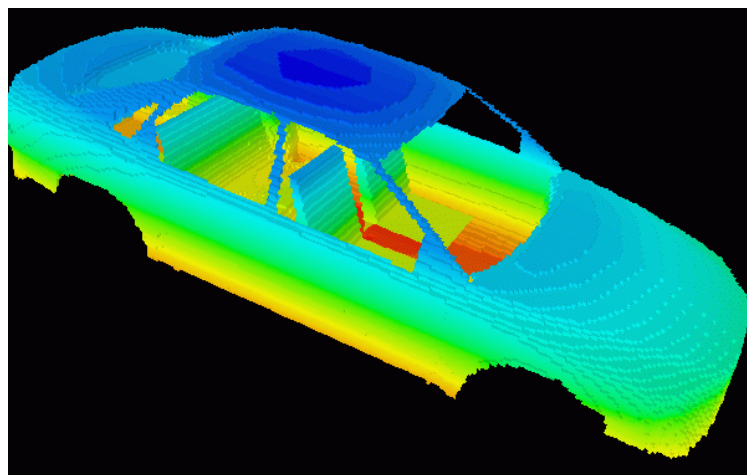
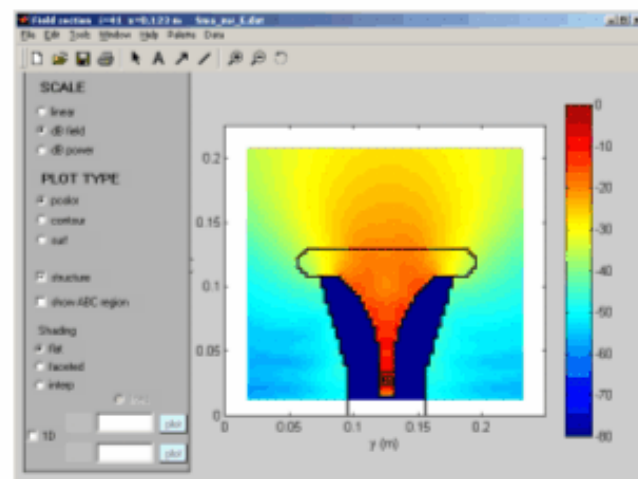
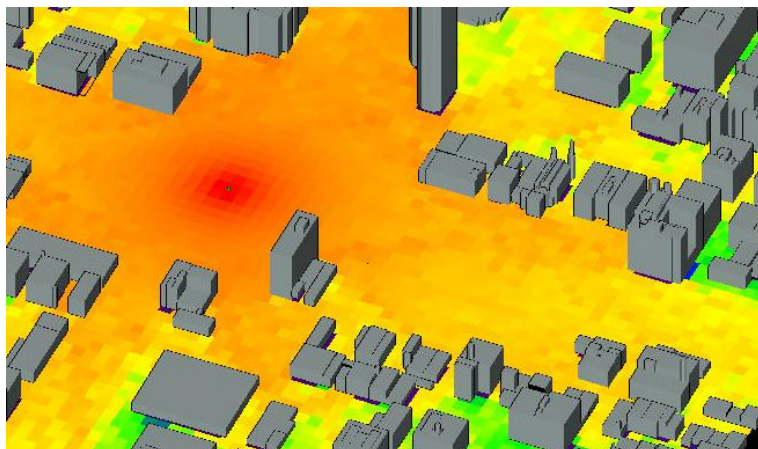


Pulsed regime



- The whole signal energy, after bouncing and absorption will leave the computation domain
- Test over global energy

Examples



Available Solvers

XFDTD, Remcom,

www.remcom.com/

EMPIRE, IMST Gmbh,

www.empire.de

Microwave Studio, CST,

www.cst.com

SEMCAD, Speag,

www.speag.com

BEST, Univ Roma Tor Vergata,

M. N. O. Sadiku, “*Numerical Techniques in Electromagnetics*”, CRC Press, Boca Raton (FL) 1992

A. S. Kunz, R. J. Lubbers, “*The Finite-Difference Time-Domain method for Electromagnetics*”, CRC Press, Boca Raton (FL) 1993

A. Taflove, “*Computational electromagnetics: The Finite-Difference Time-Domain method*”, Artech House, Norwood, MA, 1995

A. Taflove, “*Advances in Computational electromagnetics: The Finite-Difference Time-Domain method*”, Artech House, Norwood, MA, 1998